Congruences and last digits

High school · number theory · Lesson and method · 03

Prerequisites. Divisibility, powers and Euclidean division.

Key point · Ideas and method. $a \equiv b \pmod{m}$, with $m \geq 2$, means that m divides $a - b$. They have the same remainder upon division by m. For example, $-1 \equiv 6 \pmod{7}$.
Congruences are preserved by addition and multiplication, hence by positive integer powers. Work modulo 10 to find a last digit.
A factor cannot always be cancelled: $2 \times 1 \equiv 2 \times 4 \pmod{6}$, but 1 is not congruent to 4 modulo 6. A factor coprime to the modulus can be cancelled.

Example 1. The last digits of $3^1, 3^2, 3^3, 3^4$ are $3, 9, 7, 1$. The cycle repeats because $3^4 \equiv 1 \pmod{10}$. Since $2026 = 4 \times 506 + 2$, the last digit of 3^{2026} is 9.

Activity 1 · Explore. Build tables of squares modulo 3, 4 and 5. Identify impossible remainders.